EE561 Advanced Static Power Conversion · Multilevel Converters module

Potential Final Questions: Modulation and Balancing

From the joint proposal by Sefer İdacı and M. Aytunç Evlice, for M. Aytunç Evlice's talk

Two questions for a closed-book exam with the instructor's cheat sheet, each designed for about 20 minutes and 25 points, on modulation and balancing: carrier-based modulation of a three-level NPC with a common offset, and a neutral-point balancing loop. Every question asks for a derivation, a numerical result and an engineering interpretation; full solutions and a marking guide follow on the later pages. All numbers differ from the lecture examples.

Question 1 · Carrier-based PWM of a three-level NPC, and its offset

MODULATION — **25 pts**

A three-phase 3-level NPC on an 800 V DC link uses phase-disposition (PD) carriers: an upper carrier between 0 and +1 and a lower carrier between −1 and 0, in phase. Within one carrier period the normalized references are r = (0.62, 0.18, −0.80) and the phase currents are i = (40, −10, −30) A. Pole voltages are measured from the midpoint O; i_O is drawn from O.

a. For each phase, give the two levels it switches between in this carrier period and the fraction of time spent at each. [6]

b. Compute the average pole voltages $\langle v_{xO} \rangle$ and the average line voltage $\langle v_{ab} \rangle$. [4]

c. Compute the average midpoint current $\langle i_O \rangle$ in this period. [5]

d. Apply min-max injection. Give the offset r_0, the new references, the new $\langle i_O \rangle$ and the new $\langle v_{ab} \rangle$. What changed, and what did not? [6]

e. Which offset would make $\langle i_O \rangle = 0$ in this period? Is it allowed, and what does this tell you about using min-max injection and neutral-point balancing together? [4]

Question 2 · Designing a neutral-point balancing loop

BALANCING AND CONTROL — **25 pts**

A 3-level NPC with carrier PWM has a split DC link of 2 × 3.3 mF; an ideal source holds $v_{C1} + v_{C2}$ constant, so $C\, d\Delta V/dt = i_O$ with $\Delta V = v_{C1} - v_{C2}$ and i_O drawn from the midpoint. The phase currents are balanced with $\hat{I}$ = 40 A, and m = 0.9. A common offset r_0 is added to all references.

a. Starting from $\langle i_O \rangle = \Sigma_x (1 - |r_x + r_0|)\, i_x$, derive $\partial \langle i_O \rangle / \partial r_0$ and its average over a fundamental period for $r_x = m \cos \theta_x$, $i_x = \hat{I} \cos(\theta_x - \varphi)$. [6]

b. With $r_0 = K \cdot \mathrm{sign}(P) \cdot \Delta V$, derive the closed-loop time constant and choose K for τ = 20 ms at cos φ = 0.8. [6]

c. With the same K, what is τ at cos φ = 0.2? What happens in rectifier mode if sign(P) is omitted? [5]

d. At unity power factor the midpoint current contains a third harmonic of amplitude $8m\hat{I}/(5\pi)$. Estimate the peak-to-peak ripple of ΔV. Can — and should — the balancing loop remove this ripple? [8]

Solutions and marking guide

Solution 1 · Carrier-based PWM of a three-level NPC — 25 pts

(a) With PD carriers a positive reference switches between P and O, a negative one between O and N; the time away from O is $|r|$. Phase a: P 62 %, O 38 %. Phase b: P 18 %, O 82 %. Phase c: N 80 %, O 20 %. [6]

(b) $\langle v_{xO} \rangle = r_x \cdot 400$ V: +248 V, +72 V, −320 V. $\langle v_{ab} \rangle = 248 - 72 =$ **176 V**. [4]

(c) $\langle i_O \rangle = \Sigma(1 - |r_x|)\, i_x = 0.38 \cdot 40 + 0.82 \cdot (-10) + 0.20 \cdot (-30) = 15.2 - 8.2 - 6.0 =$ **+1.0 A**. [5]

(d) $r_0 = -(0.62 + (-0.80))/2 =$ **+0.09**; $r' = (0.71, 0.27, -0.71)$. $\langle i_O \rangle = 0.29 \cdot 40 + 0.73 \cdot (-10) + 0.29 \cdot (-30) = 11.6 - 7.3 - 8.7 =$ **−4.4 A**. $\langle v_{ab} \rangle = (0.71 - 0.27) \cdot 400 =$ **176 V**, unchanged. The common offset leaves every line voltage unchanged and moves the peak references away from ±1 (headroom), but it changes the midpoint current: min-max injection is not neutral for the midpoint. [6]

(e) While no reference changes sign, $\langle i_O \rangle(r_0) = 1.0 - \Sigma\, \mathrm{sign}(r_x)\, i_x \cdot r_0 = 1.0 - (40 - 10 + 30)\, r_0 = 1.0 - 60\, r_0$ (check: $r_0 = 0.09$ gives −4.4 A). Zero at $r_0 \approx$ **+0.017**: allowed, since all $|r_x + r_0|$ stay below 1 and keep their sign. Min-max and balancing act through the **same** offset and share the same headroom: the balancing loop adds its correction on top of the min-max offset, and the sum must stay within the carrier band. [4]

Solution 2 · Neutral-point balancing loop — 25 pts

(a) For an offset small enough not to change the sign of any r_x: $\partial\langle i_O \rangle / \partial r_0 = -\Sigma_x\, \mathrm{sign}(r_x)\, i_x$. Since $\langle \mathrm{sign}(\cos\theta)\cos(\theta - \varphi) \rangle = (2/\pi)\cos\varphi$ for each phase, the period average is **$-(6/\pi)\,\hat{I}\cos\varphi$**. [6]

(b) $C\, \mathrm{d}\Delta V/\mathrm{d}t = -(6/\pi)\hat{I}\cos\varphi \cdot K\, \mathrm{sign}(P)\, \Delta V = -(6/\pi)\hat{I}|\cos\varphi|\, K\, \Delta V \rightarrow$ **$\tau = \pi C/(6\hat{I}K|\cos\varphi|)$**. For τ = 20 ms: $K = \pi \cdot 3.3\times10^{-3}/(6 \cdot 40 \cdot 0.8 \cdot 0.02) =$ **2.7×10^{-3} V^{-1}** (an offset of 0.027 per 10 V of imbalance). [6]

(c) $\tau \propto 1/|\cos\varphi|$: τ = 20 ms × 0.8/0.2 = **80 ms**. Without sign(P), cos φ < 0 in rectifier mode makes the loop gain positive: ΔV grows exponentially until saturation and an overvoltage trip. [5]

(d) $\hat{I}_{O3} = 8 \cdot 0.9 \cdot 40/(5\pi) = 18.3$ A at 150 Hz. Integrating, $\Delta V_{pp} \approx 2\hat{I}_{O3}/(3\omega C) = 2 \cdot 18.3/(3 \cdot 314 \cdot 3.3\times10^{-3}) \approx$ **11.8 V** (a numerical integration gives 11.9 V). The loop bandwidth is about $1/(2\pi\tau) \approx 8$ Hz, far below 150 Hz, so it cannot remove the ripple — and it should not try: the ripple is inherent to carrier PWM with zero average, and chasing it would only inject large offsets. It is handled by sizing C (or by modified modulation). [8]